CS383 · SOFTWARE ENGINEERING

Midterm Model 2 الحل

20 درجة

محارب

01 حل Q1: MCQ

#	الجواب	السبب	المحاضرة
1	**a**	Development process.	5
2	**b**	Understandability.	5
3	**b**	Continuous feedback. الـ XP ضد التخطيط الطويل والتوثيق الثقيل.	4
4	**a**	Refactoring: تحسين الكود باستمرار بدون تغيير وظيفته.	4

Q2: Ethics حل 02

الكلمة المفتاحية	Principle	SEEPP ethic example
colleague	**COLLEAGUES**	Give a fair hearing to the opinions, concerns, or complaints of a colleague.
improve their knowledge	**SELF**	Improve their knowledge of this Code, its interpretation, and its application to their work.
informed of standards	**MANAGEMENT**	Ensure that software engineers are informed of standards before being held to them.
appropriate method	**PRODUCT**	Ensure an appropriate method is used for any project on which they work or propose to work.
conflicts of interest	**JUDGMENT**	Disclose to all concerned parties those conflicts of interest that cannot reasonably be avoided or escaped.
integrity of data	**PRODUCT**	Maintain the integrity of data, being sensitive to outdated or flawed occurrences.
responsibility for their own work	**PUBLIC**	Accept full responsibility for their own work.
prejudices	**SELF**	Not give unfair treatment to anyone because of any irrelevant prejudices.

انتبه لهذي الثنتين

جملة conflicts of interest جوابها **JUDGMENT** مو Management. وجملة integrity of data جوابها **PRODUCT** مو Profession.

03 حل Q3: FR و NFR

Functional
1. The system will allow employees to create tasks with a title, description, priority level, and due date.
2. Employees can update the status of their tasks.

Non-functional	النوع
1. The system should handle up to 100 concurrent users without delays.	Performance
2. Ensure that user data is encrypted during transmission.	Security

ويصلح كـ FR ثالث: assign tasks to team members.

04 حل Q4: الأنشطة الأربعة

1) Software specification
2) Software development
3) Software verification and validation
4) Software evolution

05 حل Q5: Development model

Project	الجواب	Justification
1	**Plan-Driven**	المتطلبات ثابتة ومعروفة من البداية، والعميل مو متوفر للـ feedback، والفرق كبيرة وموزعة وتتواصل بالتوثيق. فنحدد كل الـ specification والـ design قبل التنفيذ.
2	**Agile**	العميل مشارك باستمرار، والمتطلبات تتغير كثير، والفريق صغير (5) و co-located مع daily stand-up. الـ Agile يتقبل التغيير ويعتمد على feedback مبكر.
3	**Iterative**	كل شي متوسط: مراجعات دورية، تغيير متوسط مع core ثابت، 15 مطور، والنص يقول «regular iterations to refine the product». أكبر من اللي يناسب Agile، وفيه تغيير ما يتحمله Plan-driven.
4	**Plan-Driven**	المتطلبات ثابتة وموثقة، والمراجعة عند الـ milestones فقط، وفيه regulatory compliance يحتاج توثيق وprocess منظم.